THE SAME STUDY GUIDE · SCHOOL EDITION

The School, the Archive and the Robot

The whole project explained as if the bank were a school: the pupils, the teachers, the record books in the archive, and a robot that helps everyone understand their own files.

Same chapters as the study guide · Pupils = clients · Record books = accounts · Topics = districts · Tutors = bank staff · The robot = the LLM

Every number is the real number from the delivered version (commit fc0c281), dressed up in school words. Each chapter ends with an **In grown-up words** box that translates back.

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0 Start here 🏫

This is the same study guide, chapter for chapter, but told as a story about a school. If a grown-up word ever feels heavy, come back here: everything in the project has a school twin.

The story in one breath

Berka School keeps a **record book** for every pupil in its archive. Every time a pupil earns points, or uses them, somebody writes an **entry** in the book with the new total. The school also has **study plans** for pupils who get extra tutoring, and **badges** (junior, classic, gold).

Now the pupils and the teachers want a **robot** 🤖 they can ask questions in normal words: *"How many pupils are in each wing?"*, *"How many study plans went wrong?"*. The robot is clever with words, but it has never been inside the archive. It guesses when it is not sure, and whatever it asks the archive for, the archive hands over.

So we treated the robot as a helper who is **a bit clumsy** (it will guess) and **a bit risky** (what it asks for really happens). We tidied the paperwork, wrote a dictionary of what every word means, put a clerk at the archive desk who checks every request, and — most important — put a **lock on every drawer** that only opens for the right person. The robot never has the key.

The map: the bank and its school twin

Keep this page open the first time you read. On the left is the real project, on the right its school twin. The numbers are the real ones. One thing to know: **Berka School is a school for all ages** — the real customers are adults, so some of our "pupils" are 60 years old.

In the real project (the bank)	In the school
The bank (Berka, a Czech bank, 1993–1998)	**Berka School**
8 source files	8 boxes of paperwork from the school office
Client — 5,369	**Pupil**
Account — 4,500	**Record book**: a pupil's file in the archive, with points in, points out and a running total
Transaction — 1,056,320	**An entry** in a record book: points earned (+) or used (−), and the total after it
Balance (in CZK)	The **point total** of a record book
Disposition: owner / disponent — 5,369	Who may use a record book: the **owner** (the pupil it belongs to) or a **helper** allowed to sign in it, like a big brother
Loan — 682	A **study plan** (extra tutoring) the school gave to a record book

In the real project (the bank)	In the school
Loan status A / B / C / D	A finished and passed · B finished, **not** passed · C still running, going fine · D still running, **falling behind**
"Problematic" loan (B or D)	A study plan that **went wrong** (not passed, or falling behind)
Card — 892: junior / classic / gold	A **badge**: junior / classic / gold
Permanent order — 6,471	A standing arrangement: points moved automatically every month
Interest (`UROK`)	**Bonus points** the school adds by itself
District — 77	A **topic** (geology, maths, science …). Every pupil has a *home topic*, and every record book is filed under a topic
Region — 8	A **wing** of the school that groups topics (like the Science wing)
Statement frequency	How often a **report goes home**: monthly, weekly, or after every entry
Bank staff: relationship managers `rm_north` , `rm_south`	**Tutors** (Ms North, Mr South), each responsible for a group of pupils
Portfolio	A tutor's **group list**
`cro_global` (group risk)	The **head teacher**, whose list contains every pupil
`rm_unassigned`	A **new teacher** with no group yet
Delegation	A **substitute** or the **counsellor**: temporary access to one pupil, with a reason and an end date
`client_3763` , `client_3764`	Two **pupils** who log in to see their own book (3764 is a helper on 3763's book)
The database (PostgreSQL)	**The archive building**
The LLM (Claude Sonnet 5.5; Ollama qwen2.5-coder:3b)	**The robot** 🤖 (the big robot; the little robot)
SQL	The robot's **request slip** to the archive

The school in numbers

5,369
pupils (clients)

4,500
record books (accounts)

1,056,320
entries in the record books (transactions)

682
study plans (loans)

77
topics (districts), in 8 wings (regions)

124
checks on the paperwork (data quality rules)

1
entry put in the "do not use" tray (quarantine)

10 / 10
questions the big robot answered right (recorded run)

Two versions of the homework

The school handed in its homework on **28 October 2026** (the *delivered version*, `delivery-freeze-2026-10-28`, commit `fc0c281`). After handing it in, two things were improved on a separate page (a separate branch): how the school decides which entry was *last* on a busy day, and an extra check that the robot's request really asks for what it says it wants. In this guide those later improvements are always marked **"Later we fixed it"**.

> The one thing to remember
>
> When you talk about the homework you handed in, use the numbers you handed in. The average point total at the end of December 1998 was **43,808.94** in the delivered version. The later fix makes it 43,810.47 — that is "what I improved afterwards", never "what I handed in".

How to use this edition

1. Read this school edition first, in one go, to get the whole picture.
2. Then read the same chapter in the original study guide (01): the chapter numbers match.
3. Each chapter here ends with **In grown-up words** — the real terms and IDs (D for decisions, A for assumptions, L for lessons) so you can switch back instantly.
4. Boxes called **Oops!** are the weak spots. The full list is in the Lessons Learned document.

1 The assignment, and the school rules 📝

What the school asked for, what we built for each part, and the house rules we followed the whole way.

What the school asked for

The pupils and teachers want to ask a robot questions about the archive in normal words. A robot can only answer well if the paperwork is tidy, the words mean one thing, and every person sees only the files they are allowed to see. So the school gave us three jobs and ten test questions.

Job	In school words	What we made
1. Check the paperwork	Find the mistakes in the record books, give each kind of mistake a colour (red, orange, green) and say *why* that colour	124 checks, every one with its colour and its reason; 14 kinds of mistakes found
2. Make it robot-ready	Build a tidy filing cabinet, write a dictionary so the robot understands the words, and let the robot answer the 10 test questions	A filing cabinet, a dictionary the school checks every time, and the robot's answers next to the answer key
3. The right files for the right people	Explain how each teacher only sees their own pupils, why the robot cannot sneak past, and how a new teacher joins without rebuilding anything	Not just explained — built: a lock on every drawer, a key card plus a code, and tests that try to break in

The school also said: **"This should take one or two days. If you spend most of your time building the building instead of sorting the paperwork, you misunderstood the job."**

Oops! — we built more than a fence

We also built a key-card-and-code door, a website tour, instructions for putting it online, a 55-slide presentation and pretend robot assistants. Be ready for the question. The honest answer: the three jobs were done first; the extras were asked for and each one shows a safety idea; the building itself stayed simple (one archive, plain instructions). What we would drop first: the online instructions, the pretend assistants, then the website tour. (L-19)

Nine house rules

1. **Show, don't tell.** Every number comes from a step we actually ran, and the logbook says which. If a step did not run, it says *NOT EXECUTED* — we never fill in a guess.
2. **Photocopy first, never correct the original.** When something is fixed, the old version stays right next to it.
3. **Mark mistakes, don't hide them.** The tidying room throws nothing away; the checking room decides later.
4. **A colour must *do* something.** Red goes in the "do not use" tray, orange gets a warning sticker that travels with the answer, green is just noted.

5. **One word, one meaning, one place.** "Struggling study plan", "owner", "points in" are decided once.
6. **Don't trust a green tick.** We double-check the cabinet after every build, hide 20 fake mistakes on purpose to see if the checks catch them, and test the locks by logging in as the person who should *not* get in.
7. **The robot is a guest.** It reads the dictionary and the question, nothing else; whatever it writes is checked before it reaches the archive.
8. **Keep the answer key away from the robot.** And keep each robot's homework in its own folder.
9. **Write every big decision down**, with the other options and what it costs.

Who helped

Helper	What they did
An AI building assistant (Claude Code), directed by you	Wrote most of the work — every step run and checked against the archive
The big robot (Claude Sonnet 5.5) — the one being tested	Wrote the request slips for the 10 questions, seeing only the dictionary. No tools, no archive, no peeking
The little robot (a small free model on the laptop)	Answered the same questions through the same doors, in its own folder

Oops! — "written by hand"

Some papers say the answer key was "hand written" or "worked out by a person". The exact truth is: it was written in the same AI-assisted building session, **not by the robot being tested**, and every answer was later worked out a second time from the original paperwork. Say it that way. (L-12)

Why this archive building

We chose an archive (PostgreSQL) that can put **a lock on each drawer** (row level security). A simpler archive (DuckDB) has no drawer locks, so job 3 would have been just a story instead of something we could prove by trying to break in.

In grown-up words

Tasks 1–3 of the brief; deliverables: workbook tabs, ZIP, two-page DDR. Expected 1–2 days. Working rules = chapter 1 of the study guide (D013, D020, D004, D010, D011). Stack: PostgreSQL 16/18 (D002), SQL files + Python + Makefile, sqlglot guard, Claude CLI as a sandboxed completion (D005), natural keys (D007).

2 The archive building, room by room 🏛️

The archive has six rooms. Paperwork walks through them in order, and each room has one job and a promise to the next.

1 · Copy room

The 8 boxes of paperwork, copied exactly — not one letter changed — with a fingerprint for each box.

`raw (bronze)`

2 · Tidying room

Dates made readable, codes turned into words, + and − decided. Nothing thrown away, originals kept beside.

`stg (silver)`

3 · Checking room

124 checks with red, orange or green stickers, the "do not use" tray and the logbook.

`dq`

4 · Filing cabinet

Neat drawers: pupils, record books, study plans, entries. One meaning per word. Checked after every build.

`core (gold)`

5 · Reading shelf

The only shelf anyone may read from — 11 labelled folders, each one honouring the drawer locks.

`ai`

6 · Key office

Who is who, which group belongs to which tutor, substitutes, key cards and codes. The drawer locks ask this office.

`sec`

There is also a seventh, a **practice room** (`inject`): a copy of the tidying room with 20 fake mistakes hidden on purpose, used only to test the checks. Nothing from it may ever reach the filing cabinet.

One question, all the way through

Ms North, a tutor, asks the robot: *"What percentage of study plans went wrong?"*

1 · Dictionary
The robot gets the question and the school dictionary — nothing else.

2 · Request slip
The dictionary says "went wrong = not passed or falling behind", already marked on every plan. The robot writes a slip.

3 · Desk clerk
Checks the slip: one "please show me" request, only from the reading shelf, at most 10,000 pages.

4 · Key card + code
The slip is carried in on Ms North's own key card and code.

5 · Drawer locks
The drawers only open for her own pupils' files. The robot never sees the lock.

6 · Answer
Ms North gets her group's answer. The head teacher, who has every pupil, gets 76 of 682 = 11.14%.

Try saying it

"Three things protect the archive and they do different jobs. The dictionary stops the robot from having to guess. The desk clerk stops dangerous request slips. And the drawer locks decide which files exist for *you* — that one would hold even if the other two failed."

Three promises between the rooms

Promise	Kept?
The tidying room has exactly the same pages and points as the copy room	Yes — 11 checks, all equal
The filing cabinet = the tidying room minus the "do not use" tray	Yes — 9 checks, all equal
The dictionary matches the real cabinet (every folder and label exists)	Yes — 0 problems

In grown-up words

Schemas `raw` → `stg` → `dq` → `core` → `ai`, with `sec` beside `core`; `inject` for seeded defects. The path: prompt + context pack → model JSON → `sql_guard` (one read-only SELECT on `ai`, LIMIT ≤ 10,000) → two-factor login → `BEGIN READ ONLY`, 60 s timeout → RLS on `core.fact_loan` via the `security_invoker` view `ai.loan`. Reconciliation: raw = stg (11 checks), core = stg − quarantine (9 checks).

3 Step 1 — Copy the paperwork exactly 📦

Before anyone touches anything, make a perfect copy of what arrived, and prove it is complete.

The school office handed over **8 boxes** of paperwork: pupils, record books, who-may-use-which-book, entries, study plans, badges, standing arrangements and the list of topics. All together **1,079,680 lines**.

- We copied every line **exactly as written** — even the messy ones. Nothing was corrected here.
- Every copied line got a little label: *which box* and *which line number* it came from.
- Every box got a **fingerprint** (a SHA-256 code). If even one letter changes, the fingerprint changes, so we can always prove nothing was swapped behind our backs.
- We counted the lines in each box and compared them with the official list that comes with the dataset. All 8 matched: 4,500 record books, 5,369 pupils, 5,369 who-may-use lines, 892 badges, 682 study plans, 77 topics, 6,471 standing arrangements, 1,056,320 entries.

Why so strict? Because one day somebody will say *"that number is wrong"*. Then the first question is: what actually arrived? The copy room answers that, and nobody can argue with it.

Why not fix things straight away? If you try to turn messy text into a date while copying, a broken date either stops the whole copy or quietly disappears — and then you can never find it again. So the copy room fixes nothing, on purpose.

The practice room. The real paperwork is quite tidy, so most checks would never find anything — and a check that never finds anything proves nothing. So we made a separate practice copy with 20 mistakes hidden on purpose, to test the checks, and never mixed it with the real paperwork.

Try saying it

"The copy room is the one room nobody can argue with: here is exactly what arrived, with its fingerprint and line number."

In grown-up words

Bronze `raw`: 8 `.asc` files from the public mirror (D003), every field as text (C01), `_source_file` / `_source_row` (C02), SHA-256 per file in `raw.load_audit`, counts match the published reference. Defects injected only into schema `inject` (D004). Command `make ingest`.

4 Step 2 — Look before you sort 🔍

Before writing any checks, look at every column and write down what looks strange, with a count.

We went through every column of every list and wrote down: how many are empty, how many different values there are, the smallest, the biggest, the most common. Then we made a **list of suspicions** — before writing a single check. A few of them:

Suspicion	How many
Entries with no "from which school" information (normal for cash-like entries)	760,931
One column writes "nothing" in three different ways	535,314
Entries with no type, because they are bonus points the school adds by itself	183,114
Record books that went below zero points for a while	2,999
Pupils whose home topic is different from the topic their book is filed under	409
Books with several entries on their very last day, so "the last entry" is unclear	183
Owners who were younger than 15 when their record book was opened	114
Entries worth exactly zero points	14
Owners younger than 15 when a study plan started	6
One topic with a "?" instead of a number	1
Two entries that look exactly the same	1

Four more suspicions found **nothing** in this copy (for example, a badge belonging to a helper instead of the owner). We kept those checks anyway, because the real school's paperwork might have exactly that mistake.

Oops! — "it doesn't depend on the order"

The suspicion list says one test "does not depend on the order of entries". It does: it needs to know which entry was the *last* one on a busy day, and it guessed wrong in the same way as the filing cabinet does (chapter 7). (L-03)

In grown-up words

`make profile` writes `dq/results/profile.md` and `candidates.md` (13 candidates present, 4 absent) from staging. Rules were written from the candidates, not from a checklist.

5 Step 3 — Tidy up, but never erase 🧹

Make the paperwork readable without crossing anything out: the original always stays right next to the tidy version.

There were 21 tidying steps, and only one of them takes something away — a single entry that was written twice. Everything else changes *how it looks*, never *what was written*:

Step	What we did	How much
C03	The paperwork writes "nothing" in seven ways ('', ?, NA, N/A, NULL, -, unknown). We turned all of them into one real "empty", and kept the original next to it	1,380 lines
C04	Numbers became real numbers; if one could not be read, it became "empty" instead of stopping everything	0 failed
C05	Dates like `950103` became real dates (3 January 1995). Two-digit years mean 19xx — safe, because everything is between 1993 and 1998	1,062,394 dates
C06	Badge dates had a clock time stuck on the end (`931107 00:00:00`); we took it off	892
C07–C08	**The secret birthday code.** A pupil's number hides their birthday, and for girls the month has 50 added (May = 55). We decoded birthdays and girl/boy from it, and labelled both "worked out, not written down"	5,369 pupils
C09	Czech words became English: `POPLATEK MESICNE` → *monthly*. The Czech word stays beside the English one	2,466,175
C10	**Plus or minus, decided once.** `PRIJEM` means points coming in (+). `VYDAJ` and `VYBER` mean points going out (−). The official description doesn't even mention `VYBER` !	1,056,320 entries
C11	"Went wrong" (status B or D) is written on every study plan, once	76 plans
C12	Every change is written in a note on the line it changed	1,380 lines

Why never erase? If you correct a value and throw the old one away, a week later nobody can tell your correction from what was really written. And if you quietly throw a line away, every "how many" and every percentage changes without anyone knowing.

Why decide plus and minus only once? Every question about points coming in or going out depends on reading those Czech words right. If the robot gets `VYBER` wrong, it doesn't crash — it gives a confident, believable, **wrong** number. So the tidying room decides it once, and the running totals in the record books confirm that the choice is right.

Mark, don't hide. The tidying room never throws anything away. If it threw away the double entry, the checking room would have nothing to find, and the report would say "all clean".

Try saying it

"Twenty-one tidying steps, and exactly one removes a line — the double entry. Everything else is converted, with the original one column away. And I can prove nothing got lost: the tidy pile has exactly the same lines and points as the copy."

In grown-up words

Silver `stg` (`sql/10_stg.sql`): C03–C12 of `docs/CLEANING_LEDGER.md`. Decisions D013 (never repair in place), D014 (placeholders → NULL, original kept), D015 (19xx; failed parse = NULL), D016 (gender and birth date derived from `birth_number`), D017 (one code map; LEASING lesson), D018 (sign resolved once; `VYBER` is a debit, A29), D020 (graded, not gated). `is_problematic` is computed in both staging and core (L-18).

6 Step 4 — Traffic-light checks 🚦

Check everything, give every kind of mistake a colour, and make sure the colour actually does something.

We wrote **124 checks** (89 written by hand, each with a reason, plus 35 automatic "is anything empty?" checks). They look for empty boxes that should be filled, things written twice, entries that point to a record book that doesn't exist, impossible values, and lists that disagree with each other.

🔴 Red — do not use

The line goes into the **"do not use" tray** with a note saying why. It never reaches the filing cabinet. 1 check fired, on **1 line**.

🟠 Orange — careful

The line stays, but gets a **warning sticker**. The sticker travels all the way to the answer, so whoever reads it is warned. 6 checks fired: **440 stickers** on 352 lines.

🟢 Green — just noted

Untidy, but no answer changes. We tidy it and write it in the logbook. 12 checks fired.

On the real paperwork, **19 checks** found something, which adds up to **14 kinds of mistakes**.

The best example: same check, opposite colour

14 entries are worth **zero points**. A check saying "an entry must be more than zero" sounds obviously right. But those 14 are real: bonus-point days where the bonus worked out to nothing. Making them red would break the running totals of 14 record books — to fix nothing. So they are **green**.

An entry with **negative** points where it should be positive would be **red**, because then points move the wrong way.

Proving the checks work

- **The practice room:** 20 mistakes hidden on purpose, one for each kind of check. A test makes sure each one is caught by the check that is supposed to catch it. All 20 are caught.
- **Nothing lost:** the tidy pile has the same number of lines and the same point totals as the copy (11 comparisons). The filing cabinet equals the tidy pile minus the "do not use" tray (9 comparisons: 1,056,320 − 1 = 1,056,319 entries, and every point total equal).

Oops! — the practice mistakes almost removed real files

At one point the "do not use" tray did not say which pile a line came from. So when we ran the practice room, its fake mistakes started blocking **real** lines: 76 topics instead of 77, and 11,024 entries pointing nowhere. The cabinet checks noticed before anything was handed in. Lesson: a green tick doesn't mean it is right — check the result. (D008)

Oops! — the red check can only catch zero-point twins

The "written twice" check needs two entries with the same book, day, type, points **and the same total afterwards**. But a real double entry changes the total, so the second copy never matches. The check can only catch twins worth zero points — and the one line in the tray is a 0.00 entry. Meanwhile 1,081 groups of identical bonus-point entries on the same day have no sticker at all (157,289.70 points). Are they mistakes? Only the school office can say. (L-04)

Oops! — some orange stickers were our own fault

Of the 55 record books whose totals "didn't add up by more than 100", **41** only looked wrong because of the "who wrote last?" puzzle in chapter 7. (L-03)

In grown-up words

`dq/dq_rules.yaml` (89 rules) + 35 completeness sweep = 124 executed; 19 fire (block 1, warn 6, info 12) = 14 defect types. Quarantine 1 row (`UNQ_TRANS_BUSINESS_KEY`), 440 warn flags on 352 rows, injector 20/20 (`tests/test_dq.py`). Reconciliation 11 + 9 checks. D004, D008, A20, A30.

7 Step 5 — The filing cabinet 🗄️

Neat drawers, each with a clear rule about what one card means, and every important word decided in exactly one place.

The drawers

Drawer	One card means	Cards
Pupils	one pupil	5,369
Record books	one record book	4,500
Topics	one topic	77
Calendar	one day (1993–1998)	2,191
Badges	one badge	892
Who may use which book	one owner-or-helper link	5,369
Entries	one entry	1,056,319
Study plans	one study plan	682
Month-end totals	one record book on one last-day-of-the-month	185,615

Why "one card means one thing" matters. If the pupil drawer accidentally had a pupil twice, *every* count of pupils would be too high, and nothing would warn you. The robot would happily give a confident wrong number. So we check this every time.

Why drawers instead of one giant list? The questions are about four different things — pupils, record books, study plans, entries. In one giant list, a pupil would appear once for every entry, and every "how many pupils" would need a trick (`count distinct`) that robots often get wrong without noticing.

One word, one meaning, decided here

Word	Means
The **owner** of a record book	The one pupil the book belongs to — exactly one per book. A helper who may sign in it is *not* the owner
A study plan that **went wrong**	Status B (finished, not passed) or D (running, falling behind): 76 of 682
Points coming in	Entries that add points (`PRIJEM`), counted as positive numbers
The **month-end total**	The total after the last entry on or before the last day of the month; if there was no entry that month, last month's total is carried over

The puzzle: who wrote last? 🧩

Sometimes a record book has several entries **on the same day**, and the paperwork doesn't say at what time. So which one was last — and what was the total at the end of that day?

We decided: *the one with the highest entry number*. But entry numbers are **not** in time order. The school's bonus points get high numbers, yet they are written *before* the end-of-month fee.

There was a better clue all along: every entry says what the total was **after** it. So the next entry must start where the previous one ended — like dominoes.



That is record book **3888** on 31 January 1998. The bonus has the higher number, so our rule said "10,501.30". The dominoes say the withdrawal came last: **9,301.30**.

Oops! — our most exciting mistake was our own

Our report said this record book was "off by exactly 1,200 — an entry must be missing!". It wasn't missing at all: it was our own wrong "who was last" rule. That story is even written in the robot's dictionary. The highest number was right only **0.4%** of the times the dominoes could decide. (L-01, L-02)

Later we fixed it

After handing in the homework, we switched to the domino rule: the end of the chain is the last entry; if the dominoes can't decide, we mark the day "ambiguous". The average month-end total in December 1998 moved from 43,808.94 to 43,810.47 (+1.53), but 107 record books changed, by up to 1,200. Material "doesn't add up" stickers went from 55 to 14. (D028)

Try saying it

"The average was robust to the rule — within two points — but individual record books were not. That's why the rule had to change, and why 'we can't know the order' should have been tested instead of written down."

In grown-up words

`core` star schema: `dim_client`, `dim_account`, `dim_district`, `dim_date`, `dim_card`, `bridge_disposition`, `fact_transaction`, `fact_loan`, `fact_account_month_end_balance` (185,615 rows, 558 carried forward), curated views `v_account_owner`, `v_loan_enriched`, `v_client_region`. D007 natural keys, D019 month-end with highest-`trans_id` tie-break (A14, caveat C2), D012 rebuild restores the ai views.

8 Step 6 — The answer key, written before the test ✅

Before the robot is asked anything, write down the right answers yourself — and keep them where the robot can never see them.

- For each of the 10 test questions we wrote the right request slip ourselves, against the filing cabinet, and stored the answers. That is the **answer key**.
- Above every answer we wrote **how we understood the question** — for example "region means the wing of the pupil's *own* home topic, not the topic their book is filed under". Where a question could be understood another way, we wrote down the other way too, and how much it changes.
- The robot **never** sees the answer key, the slips, or how it will be marked. A test checks that no answer number appears anywhere in the dictionary.
- On hand-in day we ran the answer key again: all ten identical. And every answer was worked out **a second time** from the original copied paperwork, with different slips, and matched.

Oops! — who wrote the answer key?

It was written in the AI-assisted building session — not by the robot being tested, and not literally by hand. That is fine for the test (the robot never saw it), but say it exactly. The strongest argument is the second, independent calculation from the original paperwork. (L-12)

In grown-up words

`sql/60_reference_queries.sql` → `dq/results/reference_results.json`, interpretations A1–A16; `tests/test_semantic.py::test_context_pack_contains_no_answers`; independent re-derivation from `raw` text tables.

9 Step 7 — The school dictionary 📖

One book holds every word the robot is allowed to use, and the school refuses the book if it disagrees with the real filing cabinet.

What is in the dictionary

- **11 folders** on the reading shelf, with what one card means and every label explained.
- **12 official sums** ("certified metrics"), like *share of study plans that went wrong*, each with the date it uses.
- **19 links** between folders, each saying whether one card on the left can match many on the right.
- **10 words**, each with what it **is** and what it is **not**: *a helper is not the owner*; *points coming in is not "points in minus points out"*; *"Q1" in a question means January–March, not test question 1.*
- **7 warnings** with how many lines they affect (the orange stickers), which the robot must mention in its answer.
- **12 rules for the robot's request slips**, in short:

#	Rule
1	Only ask for folders on the reading shelf
2	One "please show me" request per slip
3	Never ask to change, add or delete anything
4	Never try to see more than you are allowed, and don't mention the locks
5	Use the official sums exactly as written; never invent your own "went wrong"
6	"Per month / per year" → use the calendar, so a month with nothing shows 0
7	Use the ready-made folders that already join things together
8	Give the counts next to a percentage, and mention the warnings
9	Round percentages to 2 decimals; don't round point totals
10	If the question can't be answered, **say what is missing** instead of guessing a label
11	Filter dates on the right date column, from the first day included to the first day *not* included
12	Count at the right drawer; never add up after linking to a drawer with many cards per card

The school checks the dictionary every time

Every time we rebuild, the school compares the dictionary with the real cabinet. The build **stops** if: a label in the dictionary doesn't exist, a real label is missing from the dictionary, "one card means one thing" is not true, a link has a different shape than promised, or an official sum uses a label that doesn't exist.

Why so strict? A dictionary that is out of date gives the robot confident wrong information — and the robot has no way to check.

Oops! — the dictionary told the robot where to look

The dictionary says which folder answers which test question ("Relevant to: Q1"). It doesn't give the answers, but it does point at them, so the test is easier than real life. A fair test needs questions the robot has never seen. (L-08)

Oops! — one sum said "any folder with a gender label"

The *share of boys and girls* sum didn't say which folder. Later the robot looked for a label called "gender" on the owner folder, where it is called "owner_gender" — see chapter 12. (L-07)

In grown-up words

`semantic/semantic_layer.yaml` (11 entities, 105 columns, 19 joins with measured cardinality, 12 metrics, 7 caveats, 12 query rules QR1–QR12, 10 glossary terms) → `semantic/context_pack.md` (33,662 characters at delivery). Validated by `src/semantic_context.py` on every `make semantic`.

10 Step 8 — A lock on every drawer 🔐

Every staff member and every pupil sees only the files they are allowed to see — and the lock is on the drawer itself, so the robot can't open it, whatever it writes.

Where the lock is

There are three places you could put a rule like "Ms North may only see her own pupils":

1. **In the instructions to the robot** ("please only show Ms North's pupils"). That is a *request*, not a lock — you are asking the helper to limit itself.
2. **At the archive desk**, rewriting every slip. Better, but one cleverly shaped slip gets past.
3. **On the drawers themselves.** The drawer only opens for files the person may see. ✅ This is what we built.

The lock asks: *who is holding the key card?* — not *who does the slip say it is?*. The robot only writes slips; it never has a key card. **The lock is not written on the slip, so there is nothing on the slip to remove.** Even the head teacher sees everyone only because her list contains every pupil — not because she has a master key.

Who sees how many pupils

👩‍🏫 **Head teacher**	👩‍🏫 **Ms North**	👩‍🏫 **Mr South**	🆕 **New teacher**	🧒 **Pupil 3764**
list with every pupil	tutor, north group	tutor, south group + 1 substitute pupil	no group yet	a helper on pupil 3763's book
5,369	**1,481**	**1,387**	**0**	**1**

A brand-new teacher with no group sees **nobody**, not everybody. A badly built lock usually does the opposite.

Four ways to be allowed

Way	Example	What it allows
It's yours	Pupil 3763 and her own record book	See and use her own record and her own books
You may sign in it	Big brother 3764 may sign in his sister's book	That **one book** — not his sister's personal file, not her other books
It's in your group	Ms North and the north group	**See** only — a tutor looks after pupils but doesn't spend their points
A substitute (stored with a reason and end date)	Mr South covers one of Ms North's pupils while she's away — like the counsellor helping one pupil	Only that pupil, only until the end date (about 14 days)

A new teacher joins without rebuilding anything: add her to the staff list, give her a group, make her key card, set up her code. Leaving is one change: switch her off.

Tested by trying to break in

We logged in as people who should **not** get in and tried nine tricks: pretending to be someone else, writing a "who I am" note, going round the reading shelf straight to the cabinet, giving yourself access, switching a lock off, writing into a drawer, combining folders to see more. All nine failed. Two pupils of the same tutor tried 13 ways to see each other — all failed.

Oops! — one shelf folder didn't honour the locks

Once, three folders between the reading shelf and the cabinet were missing one setting (`security_invoker`). The study-plans folder showed all 682 plans to a teacher allowed to see none, while the drawer underneath correctly showed 0. We only found it because we logged in as that teacher and counted. (D010)

Oops! — the big brother sees a bit more than promised

Our papers say a helper sees the book "and nothing else of hers". But brother 3764 can also see his sister's **badge** (junior, its date, her pupil number) and her "owner" line, because those two drawers check *the book* instead of *the person*. (L-05)

Oops! — "hidden" turned into "no badge"

When the brother looks at the study plan on that book, the owner's details are hidden — correct. But the folder then writes **"no badge"** instead of "hidden", and leaves the report frequency empty. The truth is a junior badge and monthly reports. A hidden value became a wrong value. (L-06)

In grown-up words

Seven RLS policies on the seven client-bearing core tables; every view between consumer and policy `security_invoker` (D010); identity = `session_user` (D009); no `BYPASSRLS`; grants from ownership, disposition (D023, D024), portfolio, delegation (stored, D027) in `sec.v_access_grant`; measured counts per role; 9 bypass tests in `tests/test_rls.py`. Weak spots L-05, L-06, L-11 (the "who can reach this pupil" report has a gap for pupils without a login).

11 Step 9 — Key card and secret code 🪪

Your key card (the password) alone opens nothing. You also need a code that changes every 30 seconds — and the archive itself checks it.

1. Show your **key card** (the password of your own login).
2. Type the **code** from your code app. It changes every 30 seconds and works only **once**.
3. The archive checks the code **itself**. The check only asks for the code — it already knows who is holding the key card, so nobody can type a code "for someone else".
4. If it's right, you get a **visit pass** for 30 minutes on that connection.
5. Every drawer lock also asks: *does this person have a visit pass?* No pass → **0 files**, even with the right key card.
6. Five wrong codes in a row lock your code for 15 minutes.

Why in the archive and not at the front door? If only the front door checked codes, someone who found another way into the building would skip it. The archive checks it itself, so you can prove it: right key card, no code, nothing to see.

Oops! — the "undo" trick

The archive counts wrong codes inside your own visit. Our normal door saves every attempt, so five wrong codes really do lock you out. But someone using the archive's back door with a stolen key card could press "undo" after each wrong guess, and the counter never goes up. We found this while writing the test, and wrote it down. The drawer locks are not affected: no code, no files. (L-09)

Smaller things to know

The visit pass is tied to a connection number that can be reused (L-10); the code secrets are stored so they can be read back (A44); and in the demo, the codes are shown on screen and everyone shares a demo password — fine for a demo, never for real (A46).

In grown-up words

TOTP (RFC 6238) in SQL, `sec.verify_second_factor(code)` uses `session_user`, single-use codes, 5 committed failures lock for 15 minutes, session bound to `(db_role, pg_backend_pid)` for 30 minutes, `sec.has_verified_session()` ANDed into the entitlement resolution (D026). Identifiers D025. Rollback gap = strict xfail in `tests/test_auth.py`.

12 Step 10 — The robot: clever, clumsy and a bit risky 🤖

Treat the robot like a very fast new helper who has never been inside the archive: it will guess when it isn't sure, and whatever it asks the archive for really happens.

What could go wrong

Clumsy — wrong without knowing it	Risky — its slips really run
Asks for a label that sounds right but isn't there ("gender" where it's called "owner_gender")	Asks the archive to delete or change something
Uses the wrong date (when the book was opened instead of when the study plan started)	Hides a second, sneaky request inside the first one
Links study plans to every entry, so each plan is counted hundreds of times	Peeks into back rooms where labels mean something else
Counts *how many* when the question asks *what share*	Asks for millions of pages, or for something that takes forever
Thinks "Q1" means test question 1 instead of January–March	Tries to see more pupils than the person asking may see
Makes up its own meaning of "went wrong"	

How the robot is protected against — layer by layer

Protection	What it does	What it can't do
No peeking	The robot gets only the dictionary and the question: no archive, no files, no tools, one answer	Make the robot right
The dictionary	Takes away the reasons to guess	Check that the robot used it
The desk clerk	Reads every slip properly (like reading grammar, not looking for bad words): one "please show me", only the reading shelf, at most 10,000 pages	**Check whether the slip asks the right question**
The reading rules	Read-only visit, stops after 60 seconds, the person's own key card and code	—
The drawer locks	Decide which files exist for this person, whatever the slip says	—
The archive itself	A label that doesn't exist is an error — never an answer	Catch a slip that is valid but wrong
Marking afterwards	Compares the robot's numbers with the answer key	Protect someone asking right now

Never say

"The desk clerk let it through, so the answer is right." The clerk only checks that a slip is **safe**. It does not check that it asks the **right question**.

The day the robot asked for a drawer that doesn't exist

Someone asked: *"Find male or female loans in Q1 1995."* (In school words: *how many study plans in January–March 1995 belong to boys, and how many to girls?*) The robot wrote a slip asking the **owner** folder for a label called **"gender"**.

- The desk clerk said: ✅ *safe* — one "please show me" on the reading shelf.
- The archive said: ❌ *"there is no label called gender here"*.

Boy/girl lives in the **pupil** folder. In the owner and study-plan folders the label is called **owner_gender**. The dictionary had said the sum could use "any folder with a gender label" — that was the real cause. Nothing bad happened (no wrong number, just an error). But a slip with a label that **does** exist, asking the wrong thing, would have given a believable wrong answer.

Later we fixed it

After the hand-in, the robot must also write down **what it intends**: which official sum, which dates, grouped by what. Before the archive is even opened, the clerk checks the slip against that promise: every label must exist, the dates must cover exactly January–March, no link may count plans many times. Afterwards the school works out the same sum itself, and only shows the answer if both agree. The gender question is now refused with "use owner_gender", and the right answer is 28 study plans: 18 boys' and 10 girls'. (D029)

The little robot shows what "clumsy" looks like

A small free robot on the laptop, with the same dictionary and the same doors, got 6 right, 2 half right and 2 wrong. Every mistake was a **safe** slip that the clerk correctly let through:

- **Report frequency (Q10):** it counted *how many* plans went wrong per report frequency and picked *monthly* — but the question asks for the **share**. The right answer is *after every entry*, 18.75%. That is exactly the trap the dictionary warns about.
- **New record books per month (Q6):** it linked each book to *every day* of its month, so January showed 279 new books instead of 106.
- **Age at study-plan start (Q7):** it grouped by age instead of working out the average age.

Try saying it

"The big robot with a good dictionary got all ten right. The little robot made exactly the mistakes the dictionary warns about. Every one of those slips was safe — so safety alone isn't enough; wrong answers have to be caught too."

In grown-up words

LLM as a sandboxed completion (D005, D011): no tools, no MCP, cwd outside the repo, `num_turns: 1`. `src/sql_guard.py` (sqlglot; one read-only SELECT on `ai`, no out-of-schema functions, LIMIT ≤ 10,000), `execute_guarded` (read-only, 60 s, cap + 1 rows), RLS. The observed failure: `T2.gender` on `ai.account_owner` → `column t2.gender does not exist`. Intent checker was a prototype at delivery (L-07); integrated afterwards (D029). Ollama qwen2.5-coder:3b: 6 / 2 / 2.

13 Step 11 — Marking the robot's homework 📝

Compare the robot's numbers with the answer key — the numbers, not the handwriting.

How each question is asked. The question and the dictionary go to the robot; it sends back a slip and a short explanation; the clerk checks it; it runs on the head teacher's card (her list has every pupil, so the answers can be compared with the answer key); everything is written in the logbook. If the slip is refused or can't be read, the robot gets **one more try**, with the reason. **Nobody ever fixes the robot's slip by hand.**

How it is marked. Two right answers can use different names for their columns ("pupils" or "pupil_count"), so we don't compare the slips or the names. We compare the **numbers**: every number in the answer key must appear in the robot's answer, next to the right label, within a tiny rounding difference (0.011). For the three "top" questions, the order must match too.

Oops! — getting the marking itself right

The first marking rules allowed a "close enough" of 0.01% — which would have accepted the biggest topic being wrong by 11,513 points. Now only rounding is allowed. And once, the marker said the robot was wrong about Q6 just because it wrote "1996-01" where the key said "1". Now any normal way of writing a month counts — and we wrote down that we changed this. (D011)

The results

Robot	When	Result
Big robot (Claude Sonnet 5.5) — the official run	25 October 2026, with the older dictionary	**10 out of 10** right, all on the first try, about 2 minutes, about 40 cents
Little robot (qwen2.5-coder:3b on the laptop)	27 October 2026, own folder	6 right, 2 half right, 2 wrong

Oops! — 10 out of 10 sounds better than it is

The big robot used the dictionary from **before** it was improved, and wasn't asked again with the new one (there was no big robot on the laptop). The dictionary points at which folder answers which question. There were no surprise questions, and each question was asked only once. Fair summary: *the dictionary is good enough for a strong robot on these ten questions — it doesn't yet prove it works on new ones.* (L-08)

Later we fixed it

After the hand-in, the big robot was asked again with the new dictionary and the new intent check: 10 out of 10. Four brand-new questions were added (boys and girls, and quarters of the year): 4 out of 4.

In grown-up words

`src/run_questions.py` (one retry, never hand-corrected, run as `cro_global`), `src/evaluate.py` (value-based, absolute tolerance 0.011, order checked for Q5, Q6, Q10), `dq/results/llm_results.json` (2026-10-25, `a7da25a`, 29,500-character pack), local run in `dq/results/local/`.

14 Step 12 — Showing it to the parents 👨‍👩‍👧

- **The console tour:** seven short scenes — key card only → nothing; key card + code → your own pupils; switching between people; a pupil and her tutor asking the same question and getting different, correct answers.
- **The website tour:** seven steps in a browser — copy, tidy, check, cabinet, access, login, ask the robot. You can only ask after a real login. It even works offline with the little robot.
- **One pupil, start to finish:** a story that follows one pupil from a line of raw paperwork, through tidying, into the cabinet and through the locks.

What we handed in

Item	What's in it
The workbook (11 tabs)	Kinds of mistakes, the cabinet, the questions and answers, the checks, the tidying steps, guesses, decisions, who-sees-what, the logbook
The decision paper (2 pages)	The seven biggest decisions: the options, the choice, why, the cost, and what breaks if the school becomes 100,000 times bigger
The zip file	The dictionary, all the instructions, the tests and the results
The hand-in check (`DELIVERY_AUDIT.md`)	Every claim → where its proof is → which papers repeat it

Every step writes a line in the **logbook** with the exact version it ran on. A step that never ran shows *NOT EXECUTED*.

Try saying it

"Every number in the workbook came from a step that ran and is in the logbook. A step that didn't run says NOT EXECUTED — it's never filled in."

In grown-up words

`make demo-login` (console), `make web-open` (web walkthrough, `/api/ask` requires a completed login), `make demo` (worked example). Deliverables: workbook (11 tabs), DDR (2 pages), ZIP (183 members), `docs/DELIVERY_AUDIT.md` , `run_log.jsonl` → `STATUS_REPORT.md` .

15 The ten test questions, in school words 🎯

For each question: what it means in the school, what it really asks in the bank, the answer we handed in, how we understood it, and the trap. Topic and wing names keep their real Czech names.

Q1 — How many pupils does each wing have?

Real question: How many clients do we have per region?

Wing	Pupils
south Moravia	937
north Moravia	920
central Bohemia	664
Prague	663
east Bohemia	660
north Bohemia	561
west Bohemia	515
south Bohemia	449

- **How we understood it:** the wing of each pupil's **own home topic**, counting **all 5,369** pupils — also the 869 who are only helpers on someone else's book. (A1, A2)
- **The trap:** counting record books instead of pupils (4,500), or using the topic the book is filed under. 409 owners have a different home topic from their book.
- **Watch out:** places 3–5 differ by only four pupils, so the way you count decides their order.

Say it like this

"A pupil is a person, and a person has one home topic, so I count all 5,369 by where they belong."

Q2 — How big were the study plans given out each year?

Real question: What is the total amount of loans granted per year?

Year	Plans	Total size (points)
1993	20	2,619,276
1994	101	13,379,904
1995	90	13,344,372
1996	117	18,317,676
1997	196	30,731,364
1998	158	24,869,148

- **How we understood it:** dated by the day the plan **started** (the only date a plan has); the size is the plan's own size, not "size × months". (A3)
- **Watch out:** 1993 is only half a year — the first plan started on 5 July 1993. All years together: 103,261,740 over 682 plans.

Q3 — What share of study plans went wrong?

Real question: What percentage of our loans are problematic?

76 of 682 = 11.14% (B: finished, not passed — 31 · D: running, falling behind — 45).

- **How we understood it:** "went wrong" = B or D, out of **all** plans. Written once on every plan. This definition still needs the school's OK. (A4)
- **The trap:** only counting B gives 4.55%. Counting by *size* instead of number gives 15.09% — the plans that went wrong were bigger ones.

Q4 — Average study-plan size per badge type of the book's owner

Real question: What is the average loan amount per card type of the account owner?

Owner's badge	Plans	Average size
junior	21	221,970.29
gold	16	161,880.75
no badge	512	148,864.71
classic	133	148,808.57

- **How we understood it:** the badge of the **owner** (not a helper's badge); plans whose owner has no badge are their own group, so the total stays 682. (A5, A6)
- **Think about it:** *junior* comes first, but on only 21 plans, and those owners were about 17 years old. That looks like something odd in the paperwork, not a real discovery.

Q5 — Which 10 topics had the most points coming in during 1997?

Real question: Which 10 districts had the highest total incoming transaction amount in 1997?

#	Topic	Points in, 1997
1	Hl.m. Praha	115,130,200.10
2	Karvina	28,506,437.10
3	Ostrava – mesto	27,689,271.90
4	Brno – mesto	23,095,083.40
5	Zlin	19,723,168.50
6	Frydek – Mistek	19,087,598.80
7	Olomouc	15,916,397.00
8	Kolin	14,471,603.60
9	Usti nad Orlici	14,085,167.40
10	Prachatice	12,163,147.00

- **How we understood it:** only entries that **add** points, counted as positive numbers, dated 1997, grouped by the topic the **record book** is filed under — points arrive in a book, not in a person. (A7, A8)
- **The trap:** counting `VYBER` (points going out) as coming in; using "in minus out".
- **Open question:** do the school's automatic bonus points count as "coming in"? Leaving them out keeps the same ten in the same order. (A7, awaiting the school's OK)

Q6 — How many new record books were opened each month in 1996?

Real question: How many new accounts were opened per month in 1996?

Jan 106 · Feb 104 · Mar 124 · Apr 101 · May 91 · Jun 113 · Jul 137 · Aug 133 · Sep 108 · Oct 115 · Nov 106 · Dec 125 — **1,363** in total.

- **How we understood it:** start from the **calendar**, so a month with no new books would show 0 instead of disappearing. (A9)
- **The trap:** the little robot linked each book to every day of its month and counted 279 in January.

Q7 — How old was the owner when the study plan started, per status?

Real question: What was the average age of the account owner at loan start, per loan status?

Status	Plans	Average age
A — finished, passed	203	37.07
B — finished, not passed	31	39.60
C — running, going fine	403	38.70
D — running, falling behind	45	37.00

- **How we understood it:** the **owner's exact age** on the plan's first day, with the decimals (from the secret birthday code). Six owners look younger than 15 — impossible — so we show the average with and without them. (A10, A11)
- **Small thing:** the big robot said 39.61 for B, because the reading shelf rounds every age before averaging. Close enough for the marking, but worth knowing. (L-13)

Q8 — Boys and girls among book owners with a gold badge

Real question: What is the gender split of account owners holding a gold card?

Boys 53 (60.23%) · Girls 35 (39.77%) — 88 gold-badge owners.

- **How we understood it:** owners whose **own** badge is gold; boy/girl comes from the secret birthday code, labelled "worked out". (A12, A13, A18)

Q9 — The average point total at the end of December 1998

Real question: What was the average account balance at the end of December 1998?

43,808.94 points over all **4,500** record books — 76 of them had no entry in December, so their November total was carried over.

- **How we understood it:** each book's total after its last entry on or before 31 December 1998; books with no total yet are left out, not counted as zero. (A14, A15)
- **Typical wrong answers:** forgetting the "who was last" rule creates double rows (43,508.31); only books with a December entry (44,209.53); averaging every December entry (46,908.00).
- **The puzzle:** same-day entries were decided by the highest number (chapter 7). With the domino rule (after the hand-in) it becomes 43,810.47.

Say it like this

"About 43,809 points over all 4,500 books, 76 carried over from November. I told the school the 'who was last' rule was a weak spot, and afterwards I fixed it: the average moves 1.53 points; some single books move up to 1,200."

Q10 — Which report frequency has the most study plans that went wrong?

Real question: Which statement frequency has the highest share of problematic loans?

Report goes home …	Went wrong / plans	Share
after every entry	6 / 32	**18.75%**
weekly	11 / 91	12.09%
monthly	59 / 559	10.55%

- **How we understood it:** for each report frequency, the share of **its own** plans that went wrong. (A16)
- **The trap:** "which frequency has the most plans that went wrong" gives *monthly* (59) — but that is just because monthly has the most plans. That's the little robot's mistake.
- **Think about it:** 6 out of 32 is a small group. It is the highest share we saw, but it could be luck — "worth watching", not "proven".

16 A school 100,000 times bigger 🏙️

Imagine 100,000 Berka Schools in one archive: about 100 billion entries. What would stop working, and what would we do instead?

What gets too slow	Why	What we would do
Rebuilding the cabinet	Today we rebuild every drawer from scratch each time — hours of copying files that did not change	Split drawers by month and only redo the months that changed
The 124 checks	Each check reads everything, every time	Check only the new paperwork; sample the green checks; keep the red checks exact
Month-end totals	32 billion month-end cards rebuilt from scratch	Next month's total = last month's total + this month's entries
Answering quickly	A robot's slip that reads everything takes forever; many people at once overload one archive	Ready-made summaries for the official sums, faster drawer labels for dates, and a front desk that shares connections safely
Checking the drawer locks	A tutor with millions of books makes every lock check heavy	Keep a ready list of "who may open which book", updated when something changes
Marking the robot	Ten questions are not a real exam	A big set of surprise questions, saved samples, and someone watching for strange slips

What stays the same: the lock stays on the drawers, a new teacher still joins by being added to a list, and the dictionary stays about the same size. One archive building would be too small; we would move the same rooms, checks and locks into a bigger kind of archive.

In grown-up words

DDR "What breaks at 100,000×": incremental month-partitioned facts, partition-scoped rules, a maintained month-end snapshot, marts and a `SET ROLE` gateway, a materialised `(role, account_id)` set, a hold-out evaluation. Engine: PostgreSQL with Citus, or a columnar MPP engine, or warehouse row access policies.

17 Questions people might ask 🙋

Short answers in school words. The grown-up answers are in chapter 17 of the original study guide — same numbers.

1 Why did you build so much? It was a one-or-two-day job.

The three jobs were finished first. The extras were asked for, and each shows a safety idea. The building stayed simple. If I had to drop things: the online instructions, the pretend assistants, then the website tour.

2 Why that archive?

Because it can lock each drawer, so I could prove the robot can't sneak past by actually trying to.

3 Why never correct the original paperwork?

Because a week later nobody could tell my correction from what was really written. The original stays next to the tidy version.

4 Give one red, one orange and one green.

Red: the same entry written twice — into the tray. Orange: owners younger than 15 when a plan started — kept, with a sticker. Green: zero-point bonus entries — real, so just noted.

5 Your only red line is worth zero points. What did it protect?

Honestly, nothing in points. The check can only ever catch twins worth zero. I'd add an orange check for repeated identical entries and ask the school office about the 1,081 repeated bonus entries.

6 How do you know nothing got lost?

Count everything and add up all the points in every room: the tidy pile equals the copy, and the cabinet equals the tidy pile minus the tray.

7 Where is the lock, and why can't the robot open it?

On the drawers. It asks who holds the key card. The robot only writes slips, and the lock isn't on the slip — there's nothing to remove.

8 How does a new teacher start?

Add her to the staff list, give her a group, make her key card and code. No rebuilding. Leaving is one switch.

9 Can a helper see anything of the owner?

Not her personal file. But yes, her badge and her owner line — I promised too much there. The fix is a lock that checks the person, not the book.

10 If the desk clerk lets a slip through, is the answer right?

No. The clerk only checks that it's safe. The gender question proved it: allowed by the clerk, refused by the archive.

11 Is 10 out of 10 really that good?

It's good, but narrower than it sounds: old dictionary, the dictionary points at the answers, no surprise questions, one try each. It shows the dictionary works for these ten.

12 How do you know the robot didn't just read the answers?

It never sees the archive or the answer key, and a test checks no answer number is in the dictionary.

13 What's your biggest mistake?

The "who wrote last" rule. I guessed the highest number instead of following the dominoes. It even created a fake "missing 1,200 points" story in my own report.

14 Why didn't you put the fix into the homework?

Because it changes an official answer, and an official answer should only change when the school agrees with the new rule. So I wrote it down as a known weak spot and fixed it separately.

15 Who did the work — you or the AI?

An AI helper wrote most of it while I directed and checked every step. The robot being tested is a different one, and it never saw the answer key.

16 What are you proudest of?

That my own checks caught me — twice. And that the lock sits where the robot's words can't reach it.

18 The big decisions, one line each 🧭

The 27 decisions of the handed-in homework, in school words. The full versions — with the other options and what each one cost — are in chapter 18 of the original study guide.

#	The decision, in school words
D001	Put the homework where the teacher will look first (the top of the folder)
D002	Choose an archive that can lock each drawer (and run it even without the usual container tool)
D003	Get the paperwork from a trusted public copy whose counts match the official list
D004	Hide practice mistakes in a separate practice room, never in the real paperwork
D005	Ask the robot through a sealed window: it only receives text — no tools, no archive
D006	Keep one list of Czech-word translations, and check that every copy agrees with it
D007	Use the pupils' and books' own numbers as labels; don't invent new ones
D008	The "do not use" tray must say which pile a line came from
D009	The lock is on the drawer and asks who holds the key card — not a note anyone could write
D010	Every folder between the reader and the drawer must honour the locks
D011	The robot never peeks, and its homework is marked by the numbers, not the handwriting
D012	Rebuilding the cabinet also puts the reading shelf back
D013	Never correct the original; keep the old version next to the new one
D014	Seven ways of writing "nothing" become one real "empty"; the original stays
D015	Two-digit years are 19xx; an unreadable date becomes "empty" instead of stopping everything
D016	Birthday and boy/girl are worked out from the secret code, and labelled "worked out"
D017	One translation list; the Czech word stays next to the English one; an unknown word stops the build
D018	Plus or minus is decided once, in the tidying room
D019	The month-end total is worked out once, with the "who was last" rule written down
D020	The tidying room throws nothing away; the checking room decides
D021	A pupil who logs in *is* one pupil — not a teacher with a group of one
D022	*(Replaced by D023)* At first, a pupil only saw the books they own
D023	A pupil sees every book they are allowed to use, not only their own
D024	Being allowed to sign in a book gives you that book — never the person
D025	Three names for a pupil: an inside number, a code safe to share, and the book number they recognise
D026	The secret code is checked by the archive itself, not just at the front door
D027	Access is worked out live from the lists; only a human decision (a substitute) is stored

#	The decision, in school words
D028	*Later:* the domino rule decides which entry was last
D029	*Later:* the robot must say what it intends, and its slip must match that

19 The guesses we had to make 🤔

When the paperwork or the question didn't say, we chose — and wrote the choice down, with what changes if it's wrong. Here are all 46, in school words.

How we understood the questions

#	The guess
A1	"Wing" means the wing of the pupil's **own** home topic
A2	Count **all** pupils, also helpers who own no book
A3	A plan is dated by its start day; its size is its own size
A4	"Went wrong" = B or D — **needs the school's OK**
A5	The owner's badge means the owner's own badge, not a helper's
A6	Owners without a badge are shown as "no badge", not left out
A7	"Points coming in" = entries that add points, counted positive (bonus points included — **needs the school's OK**)
A8	Q5 uses the topic the book is filed under
A9	Show all twelve months, even a month with zero
A10	Age is exact, with decimals
A11	The owner is the owner of the plan's book; impossible ages are shown both ways
A12	Gold-badge owners are owners whose own badge is gold
A13	Count owner lines (one per book), with the number of different pupils beside it
A14	The month-end total is after the last entry on or before that day; same-day ties go to the highest number (the weak spot)
A15	Books with no total yet are left out, not counted as zero
A16	The share is worked out inside each report frequency

The paperwork, the archive and the tidying

#	The guess
A17	Two-digit years are 19xx
A18	Girls' birth months have 50 added; that is the only source of boy/girl
A19	All points are the same kind (in the bank: all amounts are Czech crowns)
A20	Points have one decimal, so totals are checked with a small rounding margin (1 point, and 100 points for "serious")
A21	A tutor sees the pupil, the books that pupil owns, and everything in them
A22	The topic list, the calendar and the translation list are open to everyone
A23	The public copy is the real paperwork (no copy from the school office arrived)
A24	The robot is a guest who never sees the archive; its slips are always checked
A25	The seven ways of writing "nothing" are the full list
A26	"Empty" after tidying means "never written down", never "we lost it"
A27	An entry with no type is a bonus point the school added itself
A28	A code we don't know stops the build instead of being guessed
A29	Only `PRIJEM` adds points; `VYDAJ` and `VYBER` take points away
A30	The book's own written total is believed; we never recalculate it
A31	The paperwork is one snapshot, with no history
A32	Everything happens between 1 January 1993 and 31 December 1998
A33	The order of lines in a box means nothing

Logging in and who may see what

#	The guess
A34	A pupil logs in with their own login, never a shared one
A35	The school office decides which login belongs to which pupil
A36	*(Replaced by A39)* A pupil only sees the books they own
A37	The topic list is not secret, so a pupil may see all 77 topics
A38	A "who read what" logbook is not needed for this homework
A39	A pupil sees every book they are allowed to use
A40	Being a helper gives you the book only — not the owner's file, not their other books
A41	A tutor's group means "see", never "spend"
A42	The share-safe code (`BRK-…`) is safe enough to show on screen
A43	A 30-second code is good enough for this demonstration
A44	The code secrets are stored so the archive can read them back (needed to check codes)
A45	Tying a visit pass to the connection number is enough to stop copying it
A46	Showing the code on screen in the demo is fine — for a demo
A47	*Later:* the boy/girl of a study plan is the owner's — **needs the school's OK**
A48	*Later:* "Q1" means January–March (calendar quarters), one row per period — **needs the school's OK**
A49	*Later:* bonus points count as points coming in — **needs the school's OK**

20 Word list 📚

School word	Real word	Meaning
Berka School	the bank	Where all the paperwork comes from
Pupil	client	A person; 5,369 of them
Record book	account	A pupil's file: points in, points out, total; 4,500
Entry	transaction	One line in a record book; 1,056,320
Point total	balance	How many points the book holds right now
Owner / helper	owner / disponent (disposition)	Whose book it is / who else may sign in it
Study plan	loan	Extra tutoring given to a book; 682
Went wrong	problematic (status B or D)	Not passed, or falling behind
Badge	card	junior, classic or gold; 892
Topic / wing	district / region	77 topics in 8 wings
Report frequency	statement frequency	monthly, weekly, after every entry
Bonus points	interest (`UROK`)	Points the school adds by itself
Tutor, group list	relationship manager, portfolio	A staff member and the pupils they look after
Head teacher	`cro_global`	Sees everyone, because her list has everyone on it
Substitute / counsellor	delegation	Temporary access to one pupil, with a reason and an end date
Copy room	`raw` (bronze)	Paperwork exactly as it arrived
Tidying room	`stg` (silver)	Readable, decoded, nothing thrown away
Checking room	`dq`	Checks, stickers, the "do not use" tray
Red / orange / green	block / warn / info	Into the tray / sticker that travels / just noted
Filing cabinet	`core` (gold)	The neat drawers, one meaning per word
Reading shelf	`ai`	The only place anyone may read from
Key office	`sec`	Who is who and who may see what
Drawer lock	row level security	Opens only for files this person may see
Key card + code	password + TOTP (two-factor login)	Both needed; the archive checks the code itself
School dictionary	semantic layer / context pack	Every word the robot may use, and what it is *not*
Robot 🤖	the LLM (Claude Sonnet 5.5; qwen2.5-coder:3b)	Writes request slips from questions in normal words

School word	Real word	Meaning
Request slip	SQL query	What the robot asks the archive for
Desk clerk	SQL guard	Checks a slip is safe — not that it is right
Answer key	reference SQL and results	The right answers, written before the test
Marking	evaluation	Comparing the robot's numbers with the answer key
Dominoes	the balance chain	Each entry starts where the previous one ended
Logbook	run log	Which step ran, when, on which version